Jacqueline R. Thompson, Ph.D.

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Education

SUNY Upstate Medical Univsersity Aug 2019 – April 2024
Doctor of Philosophy in Neuroscience *Syracuse, NY*

University of Portland Aug 2012 – Aug 2015
Bachelor of Science in Biology (Cum Laude) *Portland, OR*

Research Experience

Johns Hopkins University (Dept. Biostatistics) May 2024 – Present
Postdoctoral Researcher — Advisor: Dr. Stephanie Hicks *Baltimore, MD*

- Led analysis of 119-donor case-control spatial transcriptomic study (500,00+ spots) to identify sex-specific disease mechanisms in mood disorders. Implemented GRN inference methods to organize 500+ significant results that were integrated with donor genotypes via eQTL and colocalization analysis.
- Developed computational methods to address systematic technical artifacts in NGS platforms and mentored graduate students in releasing them as R/Bioconductor packages: *BatchSVG*, published and adopted in a 160-donor clinical pipeline, and *SpatialArtifacts*, under review at *Nature Communications*.
- Assumed leadership of a human hippocampus spatial transcriptomic atlas integrating multi-model and cross-species NGS datasets, elevating the project to publication in *Nature Neuroscience*.

SUNY Upstate Medical University (Dept. Neuroscience) Aug 2019 – April 2024
Graduate Research Assistant — Advisor: Dr. Yingxi Lin *Syracuse, NY*

- Developed a multi-task neural network model that accurately predicted developmental neuronal activity from sc/snRNA-seq expression, enabling the assessment of activity-associated changes in chromatin accessibility in paired snATAC-seq data.
- Led end-to-end analysis of several complex NGS datasets that identified a candidate signaling pathway implicated in experience-dependent maturation of mouse dentate gyrus granule cells.

Oregon National Primate Research Center June 2015 – Aug 2019
Research Assistant I-II — Advisor: Dr. Elinor Sullivan *Beaverton, OR*

- Led longitudinal analysis in a nonhuman primate model investigating the relationship between perinatal diet, maternal metabolic state and inflammation on long-term offspring neurobehavioral outcomes.

Technical Skills

Next-generation sequencing: spatial transcriptomics, sc/snRNA-seq, snATAC-seq, long-read RNA-seq, WGS
Computational Tools: R, Python, Unix/Linux, Git/Github, HPC/Slurm, Azure; SingleCellExperiment, Seurat, Scanpy, ArchR, MACS2, limma, edgeR, DESeq2; scikit-learn, PyTorch, TensorFlow; tidyverse, ggplot2, NumPy, pandas, Matplotlib
Concepts: Pipeline development, quality control, differential expression, statistical modeling, clustering, enrichment analysis, machine learning, multi-omic and cross-species integration, gene regulatory networks, data visualization

Publications

[1] Jiali Harriet He, **Jacqueline R. Thompson**, Michael Totty, and Stephanie C. Hicks. SpatialArtifacts: a computational framework for tissue artifact detection in spatial transcriptomics data. *bioRxiv*, page 2026.05.15.725260, May 2026. Revision submitted to *Nature Communications* Sept 2026.

[2] Kinnary Shah, Christine Hou, **Jacqueline R. Thompson**, and Stephanie C. Hicks. BatchSVG: identifying batch-biased genes in the application of spatially variable gene detection. *Bioinformatics*, 42(7):btag522, July 2026.

[3] **Jacqueline R. Thompson***, Erik D. Nelson*, Madhavi Tippani*, Anthony D. Ramnauth, Heena R. Divecha, Ryan A. Miller, Nicholas J. Eagles, Elizabeth A. Pattie, Sang Ho Kwon, Svitlana V. Bach, Uma M. Kaipa, Jianing Yao, Christine Hou, Joel E. Kleinman, Leonardo Collado-Torres, Shizhong Han, Kristen R. Maynard, Thomas M. Hyde, Keri Martinowich, Stephanie C. Page, and Stephanie C. Hicks. An integrated single-nucleus and spatial transcriptomics atlas reveals the molecular landscape of the human hippocampus. *Nature Neuroscience*, 28(9):1990–2004, September 2025. *These authors contributed equally to the work.

[4] Anthony D. Ramnauth, Madhavi Tippani, Heena R. Divecha, Alexis R. Papariello, Ryan A. Miller, Erik D. Nelson, **Jacqueline R. Thompson**, Elizabeth A. Pattie, Joel E. Kleinman, Kristen R. Maynard, Leonardo Collado-Torres, Thomas M. Hyde, Keri Martinowich, Stephanie C. Hicks, and Stephanie C. Page. Spatiotemporal analysis of gene expression in the human dentate gyrus reveals age-associated changes in cellular maturation and neuroinflammation. *Cell Reports*, 44(2), February 2025.

[5] Jian Xue, Andrew T. Brawner*, **Jacqueline R. Thompson***, Tushar D. Yelhekar, Kyra T. Newmaster, Qiang Qiu, Yonatan A. Cooper, C. Ron Yu, Yasir H. Ahmed-Braima, Yongsoo Kim, and Yingxi Lin. Spatiotemporal Mapping and Molecular Basis of Whole-brain Circuit Maturation. *bioRxiv*, page 2024.01.03.572456, January 2024. *These authors contributed equally to the work. Revision submitted to *Neuron* Oct 2026.

[6] **Jacqueline R. Thompson** and Yingxi Lin. Nature and Nurture Converge in the Nucleus to Regulate Activity-Dependent Neuronal Development. In Ramendra N. Saha and Serena M. Dudek, editors, *Transcriptional Regulation by Neuronal Activity: To the Nucleus and Back*, pages 301–347. Springer Nature Switzerland, Cham, 2024.

[7] Samantha Papadakis, **Jacqueline R. Thompson**, Eric Feczko, Oscar Miranda-Dominguez, Geoffrey A. Dunn, Matthew Selby, AJ Mitchell, Elinor L. Sullivan, and Damien A. Fair. Perinatal Western-style diet exposure associated with decreased microglial counts throughout the arcuate nucleus of the hypothalamus in Japanese macaques. *Journal of Neurophysiology*, 131(2):241–260, February 2024.

[8] Geoffrey A. Dunn*, **Jacqueline R. Thompson***, A. J. Mitchell, Samantha Papadakis, Matthew Selby, Damien Fair, Hanna C. Gustafsson, and Elinor L. Sullivan. Perinatal Western-style diet alters serotonergic neurons in the macaque raphe nuclei. *Frontiers in Neuroscience*, 16, January 2023. *These authors contributed equally to the work.

[9] Julian S. B. Ramirez, Alice M. Graham, **Jacqueline R. Thompson**, Jennifer Y Zhu, Darrick Sturgeon, Jennifer L. Bagley, Elina Thomas, Samantha Papadakis, Muhammed Bah, Anders Perrone, Eric Earl, Oscar Miranda-Dominguez, Eric Feczko, Eric J. Fombonne, David G Amaral, Joel T Nigg, Elinor L. Sullivan, and Damien A. Fair. Maternal Interleukin-6 Is Associated With Macaque Offspring Amygdala Development and Behavior. *Cerebral Cortex*, 30(3):1573–1585, March 2020.

[10] Madison DeCapo*, **Jacqueline R. Thompson***, Geoffrey Dunn, and Elinor L. Sullivan. Perinatal Nutrition and Programmed Risk for Neuropsychiatric Disorders: A Focus on Animal Models. *Biological Psychiatry*, 85(2):122–134, January 2019. *These authors contributed equally to the work.

[11] **Jacqueline R. Thompson**, Hanna C. Gustafsson, Madison DeCapo, Diana L. Takahashi, Jennifer L. Bagley, Tyler A. Dean, Paul Kievit, Damien A. Fair, and Elinor L. Sullivan. Maternal Diet, Metabolic State, and Inflammatory Response Exert Unique and Long-Lasting Influences on Offspring Behavior in Non-Human Primates. *Frontiers in Endocrinology*, 9, April 2018.

[12] **Jacqueline R. Thompson**, Jeanette C. Valleau, Ashley N. Barling, Juliana G. Franco, Madison DeCapo, Jennifer L. Bagley, and Elinor L. Sullivan. Exposure to a High-Fat Diet during Early Development Programs Behavior and Impairs the Central Serotonergic System in Juvenile Non-Human Primates. *Frontiers in Endocrinology*, 8, July 2017.

Oral Presentations

"Sex-specific spatiomolecular evidence of cell stress and neuronal disequilibrium across mood disorders in the human prefrontal cortex". June 30, 2026. Lieber Institute for Brain Development Internal Seminar Series. Baltimore, MD.

"Getting more out of NGS: From reporter mouse models to psychiatric case-control studies". December 16, 2025. Invited speaker at One Neuro Initiative Postdoc Talks. Baltimore, MD.

"Predicting the activity status of single nuclei". March 31, 2023. Invited participant at SUNY Upstate Medical University Neuro Research Day 2023. Syracuse, NY.

"How Neuronal Communication Shapes the Brain: Single cell analyses reveal a novel gene network driving brain maturation". April 6, 2022. Selected participant at SUNY Upstate Medical University Student Research Day 2022. Syracuse, NY.

"Aberrant behavioral regulation in juvenile Japanese macaques is associated with HGF-SF, perinatal diet, and gestational environment". April 13, 2018. Selected participant at Society for Neuroscience Oregon and Southwest Washington Chapter Meeting 2018. Troutdale, OR.

"Perinatal influences of offspring behavior and neurodevelopment in Japanese macaques". March 5, 2018. Invited participant for OHSU Department of Behavioral Neuroscience Seminar. Portland, OR.

"Developmental exposure to high-fat diet programs neural development and anxiety behaviors in juvenile non-human primates". September 21, 2017. Institutional invited participant at 2017 Annual ONPRC Scientific Symposium. Beaverton, OR.

Poster Presentations

Jacqueline R. Thompson, Jian Xue, Yanli Zhang-James, Yingxi Lin. Single cell RNA sequencing and machine learning reveal novel mechanisms driving activity-dependent neuronal maturation. November 9-12, 2022. Presented at Cold Spring Harbor Biological Data Science Meeting. Cold Spring Harbor, NY.

Jacqueline R. Thompson, A.J. Mitchell, Samantha Papadakis, Damien A. Fair, Elinor L. Sullivan. Probing the Diversity of Raphe Serotonergic Nuclei: The Impact of Perinatal Western-Style Diet. April 12, 2019. Presented at Society for Neuroscience Oregon and Southwest Washington Chapter Meeting 2019. Troutdale, OR.

Awards & Honors

Best Oral Research Presentation Award April 2022
Charles Ross Memorial Student Research Day 2022 *Syracuse, NY*

Best Student Oral Presentation Award April 2018
Society for Neuroscience Oregon and Southwest Washington Chapter Meeting *Troutdale, OR*

Arthur A. Schulte Jr. Scholars Award 2012 - 2015
University of Portland Merit Scholarship *Portland, OR*

Teaching Experience

Data Analysis in R for Neuroscientists
1/2 CR. Fall, 2022. SUNY Upstate Medical University. Syracuse, NY.

- Created a new course for biomedical graduate students interested in gaining coding experience. Designed the syllabus and material and instructed 8 lectures covering the basics of R utility for data wrangling, visualization, and simple statistical analysis.